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### A Two-Way Mixed ANOVA (2B)

In this study, cholesterol dataset were compared between two margarine brands (A and B) over three time points: Before, After 4 weeks, and After 8 weeks. The dependent variable was cholesterol level, while the independent variables were margarine brand and the time.

**Dependent variable:** Cholesterol level

**Independent variables:** Margarine and the time

#### Assumptions

**Assumption #1: The dependent variable, which is the cholesterol level, is measured at the continuous level.**

This condition was met because the cholesterol dataset is in numerical form or quantitative measurement.

**Assumption #2: Independent variable**

The margarine has two groups (A and B), and the time has three repeated measurements (Before, After 4 weeks, and After 8 weeks).

**Assumption #3: Independence of observation is assumed between the participants**

This is satisfied because the participants in the dataset in one group were different from the participants in the other group.

**Assumption #4: There were no major outliers that changed the pattern of the data**

There were no extreme outliers that would change the overall interpretation of the results.

**Assumption #5: The dependent variable was approximately normally distributed within each margarine brand and the time, as assessed by the Shapiro-Wilk test.**

(Insert Table from Jamovi)

**Assumption #6: Homogeneity of variance was checked using Levene's test at each time.**

#### Assumptions

| Homogeneity of Variances Test (Levene's) |      |     |     |      |
|------------------------------------------|------|-----|-----|------|
|                                          | F    | df1 | df2 | p    |
| Before                                   | 5.03 | 1   | 16  | .039 |
| After4weeks                              | 5.42 | 1   | 16  | .033 |
| After8weeks                              | 5.98 | 1   | 16  | .026 |

Levene's test was significant at all three times, which means that the variance across the two margarine brands was not equal.

## Estimated Marginal Means

### Time \* Margarine

Estimated Marginal Means - Time \* Margarine

| Margarine | Time        | Mean | SE    | 95% Confidence Interval |       |
|-----------|-------------|------|-------|-------------------------|-------|
|           |             |      |       | Lower                   | Upper |
| A         | Before      | 5.95 | 0.405 | 5.09                    | 6.80  |
|           | After4weeks | 5.47 | 0.390 | 4.64                    | 6.30  |
|           | After8weeks | 5.41 | 0.382 | 4.60                    | 6.22  |
| B         | Before      | 6.78 | 0.363 | 6.01                    | 7.55  |
|           | After4weeks | 6.14 | 0.349 | 5.40                    | 6.88  |
|           | After8weeks | 6.08 | 0.342 | 5.35                    | 6.80  |

### Group Summary

| Margarine | N  | Excluded |
|-----------|----|----------|
| A         | 8  | 0        |
| B         | 10 | 0        |

## Descriptives

### Descriptives

|             | Margarine | N  | Missing | Mean | Median | SD    | Minimum | Maximum |
|-------------|-----------|----|---------|------|--------|-------|---------|---------|
| Before      | A         | 8  | 0       | 5.95 | 5.65   | 1.428 | 3.91    | 7.67    |
|             | B         | 10 | 0       | 6.78 | 6.66   | 0.866 | 5.73    | 8.43    |
| After4weeks | A         | 8  | 0       | 5.47 | 5.09   | 1.388 | 3.70    | 7.12    |
|             | B         | 10 | 0       | 6.14 | 5.99   | 0.815 | 5.13    | 7.71    |
| After8weeks | A         | 8  | 0       | 5.41 | 5.09   | 1.374 | 3.66    | 7.05    |
|             | B         | 10 | 0       | 6.08 | 5.86   | 0.779 | 5.17    | 7.67    |

Brand A started with a mean cholesterol level of 5.945 and it slowly went down to 5.479 after 4 weeks and after 8 weeks it was approximately about 5.409. It shows in the table that the Brand B also decreased, from 6.778 before went down to 6.140 after 4 weeks and 6.075 after 8 weeks. Both A and B showed a downward pattern over time.

## Repeated Measures ANOVA

### Within Subjects Effects

|                  | Sum of Squares | df | Mean Square | F      | p      | $\eta^2_p$ |
|------------------|----------------|----|-------------|--------|--------|------------|
| Time             | 4.1377         | 2  | 2.06883     | 249.01 | < .001 | 0.940      |
| Time * Margarine | 0.0800         | 2  | 0.04000     | 4.81   | .015   | 0.231      |
| Residual         | 0.2659         | 32 | 0.00831     |        |        |            |

Note. Type 3 Sums of Squares

[3]

### Between Subjects Effects

|           | Sum of Squares | df | Mean Square | F    | p    | $\eta^2_p$ |
|-----------|----------------|----|-------------|------|------|------------|
| Margarine | 6.98           | 1  | 6.98        | 1.90 | .187 | 0.106      |
| Residual  | 58.89          | 16 | 3.68        |      |      |            |

Note. Type 3 Sums of Squares

The effect of margarine was not significant,

$F(1, 16) = 1.896$  or  $1.90$ ,  $p = 0.187$ ,  $partial\ n^2 = 0.106$ .

While the effect of time was significant,  $F(2, 32) = 249.01$ ,  $p < .001$ ,  $partial\ n^2 = 0.940$ .

Also, the interaction between margarine and time was also significant,

$F(2, 32) = 4.814$ ,  $p = 0.015$ ,  $partial\ n^2 = 0.231$ .

There was a significant interaction between margarine and time on cholesterol, which means that the effect of time was not exactly the same for both A and B.

## Post Hoc Tests

Post Hoc Comparisons - Time

| Comparison  |               | Mean Difference | SE     | df   | t     | Ptukey | Pbonferroni |
|-------------|---------------|-----------------|--------|------|-------|--------|-------------|
| Time        | Time          |                 |        |      |       |        |             |
| Before      | - After4weeks | 0.5571          | 0.0322 | 16.0 | 17.30 | < .001 | < .001      |
|             | - After8weeks | 0.6196          | 0.0383 | 16.0 | 16.16 | < .001 | < .001      |
| After4weeks | - After8weeks | 0.0625          | 0.0172 | 16.0 | 3.63  | .006   | .007        |

Both A and B at each time point were not significant.

### APA Report

A two-way repeated measures ANOVA was conducted to examine the effects of time (before, after 4 weeks, and after 8 weeks) and the margarine type (A and B) on cholesterol levels. Based on the results presented, it showed significant main effect of time,

$F(2, 32) = 249.01, p < .001, partial\eta^2 = 0.940$ . This indicates that the cholesterol levels significantly changed over time. The cholesterol levels decreased after 4 weeks and continued to go down after 8 weeks.

However, there was a significant interaction between time and margarine,

$F(2, 32) = 4.81, p = 0.015, partial\eta^2 = 0.231$ . In this pattern, it has been observed that there's a pattern of change in cholesterol levels over time that differs depending on the type of margarine used.

Post hoc comparisons using Bonferroni showed that for Brand A, cholesterol significantly decreased after 4 weeks ( $p < 0.001$ ) and after 8 weeks ( $p < 0.001$ ). Similarly for Brand B, cholesterol levels also showed a decreasing trend over time.